The What, the How and the Why

The Strategy is the Journey

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August 21, 2025

1 The What, the whaT and the wHat

A note before you read on. This is not a paper; it's a journey. An observational experiment. A threshold of discovery. If you want recipes, how-to's, a flashy quick fixe to revolutionize education, stop here.

You have to embrace the journey head-on. You have to accept the challenge.

Now, if you are serious about it all, take a moment to look at the Tables of Contents of your favorite math textbooks or learning platforms.

Khan Academy. Amplify. Mathigon. You name it. Maybe take your favorite or dreaded math textbook from school or college. Do that.

Scroll through their sequence. Read the chapter titles.

Now pause — and go through the lists again. What do you notice? What pattern emerges? Pause and think.

You are at a crossroad here. Think about the Table of Contents you just read. Reconsider it. Let the whole thing speak to you. Break it apart, re-arrange it, print it, make your own, whatever.

Before you move on.

1.1 How I see it

Fractions. Linear functions. Exponential growth. Congruence. Limits. Almost all sites or books, regardless of style, audience, or pedagogy, organize learning by **concepts**.

Some platforms break them into skills, others into lessons, but the backbone remains: a progression of concepts, increasingly complex.

Now stop reading. Pick a math problem, from anywhere. Try to solve it.

Whether you succeed or get stuck, now do one thing: *look back*. What did you actually do?

What was your solving path?

Did you try a special case? Draw something? Look for a pattern? Write an equation? Make a mistake and pivot?

Now stop reading. Pick another problem. Any one. From anywhere. Everyday life, culture, economics, housekeeping, relationships, health, you name it. Try to solve it. And sketch the solving path you took.

Another problem. Change field if you like.

Now take those solving paths and lay them side by side.

What kind of moves are you making? What kind of thinking recurs? What would you call these paths?

Now compare them to the Table of Contents that you saw earlier.

Do they align?

Do they even speak the same language?

Is what you did — the human experience of solving — reflected in how we design learning?

Or is there a gap?

2 Wait but ... How ?

Let's put it this way:

*We don't **just** follow a path of concepts when we solve math problems.*

We follow **strategies** — knowingly or not.

We don't **just** say, "Let me now apply the linearity of expectation." We say:

- "Let's try a simpler case."
- "What if I draw a diagram?"
- "Can I guess and check?"
- "What if I work backwards?"
- "Maybe I can find a pattern."

These are not static concepts. They are *moves* — mental gestures, ways of approaching, ways of seeing.

They come before — or at least alongside — any formal idea is named. They are the hidden engine behind every breakthrough, every "aha," every quiet step forward in solving.

They are more general, transferable, creative. Human in the most ancient sense.

But they almost never appear in curricula as the **primary structure**.

Instead, they're relegated to the margins — a sidebar, a handout, a paragraph buried somewhere.

We first teach concepts and then move (sometimes!) to explicitly figuring out strategies.

But the reverse is more natural, more powerful, and more human:

We should teach through strategies, and let concepts emerge as landmarks along the journey.

3 Welcome to Strategy-Driven Learning

If strategies are what we actually use when solving, then why don't we design learning that way?

Imagine a curriculum built not as a staircase of concepts, but as a web of **strategies**.

Each unit wouldn't be called "Linear Equations" or "Quadratic Functions." It would be called:

- Working Backwards
- Drawing to See
- Finding and Breaking Patterns
- Guessing Strategically
- Changing Perspective
- Organizing Information
- Building from Extremes
- Undoing and Reversing
- Acting Out the Problem
- Making It Simpler
- Invent your own Strategy
- Etc.

In each of these units, students would not just practice a technique — they'd build a habit of thinking. They'd encounter problems from across mathematics — geometry, algebra, arithmetic, logic — all chosen because they *elicit* the same strategic approach.

Concepts would appear, of course. But as second steps in the journey. As small and *meaningful* discoveries.

The strategy comes first. The concept comes as a friend along the way.

A Tale of Two Structures

Concept-Based Curriculum	**Strategy-Based Curriculum**
Chapter 1: Fractions	Step [–]: Comparing Quantities
Chapter 2: Decimals	Step [–]: Estimating and Approximating
Chapter 3: Ratios and Proportions	Step [–]: Scaling Up and Down
Chapter 4: Linear Equations	Step [–]: Undoing and Reversing
Chapter 5: Area and Perimeter	Step [–]: Drawing to See
Chapter 6: Word Problems	Step [–]: Acting It Out
Chapter 7: Probability	Step [–]: Organizing Information
Chapter 8: Graphs and Functions	Step [–]: Looking for Patterns
Chapter 9: Systems of Equations	Step [–]: Working Backwards
Chapter 10: Test Preparation	Step [–]: Strategy Mixing and Reflection

Note: Steps are intentionally left without numbers. These pairings are not one-to-one translations. The strategy-based structure doesn't mirror the concept chapters — it cuts across them. A single strategy might touch many concepts. A single concept might appear under several strategies.

The point isn't to rearrange the labels – it's to reimagine the spine.

4 A New Beginning

This paper isn't about switching labels, but recovering something we lost.

Somewhere along the way, we began mistaking the *content of thinking* for *thinking itself*.

The question is: How do we think? Or, put differently: let's think about our thinking first. We designed learning to follow content, not cognition. We mapped the terrain, but ignored how people walk through it. One of the worst side-effects of this approach is that Math Teaching has been frequently degraded to teaching procedures. Wonder why students hate this fake-math?

This is a call to rebuild the path.

Not to discard concepts — but to let them emerge from action.

Not to abandon curriculum — but to reimagine its backbone.

Because if we want learners to think mathematically, or more broadly strategically!, we must show them how strategic thinkers actually walk their paths.

And that means building our lessons not just around what must be known, but around the strategies that help us know it.

Especially for those who have been left behind by concept-first schooling, who were told they "just don't get math".

Face a problem, make a strategy, test it, and refine.

Thank you for reading![1].

[1]Before we go. Still wondering about the Why? Stay tuned — more is on the way!